

AI Hybrid Based Plagiarism Detection System Creation

A Jauhar Fuad
info.ajauharfuad@gmail.com

Amar Kukuh Wicaksono
ammar@uit-lirboyo.ac.id,

M. Auzai Aqib
aqibmohammad93@gmail.com

M. Arif Khoiruddin
arif@uit-lirboyo.ac.id,

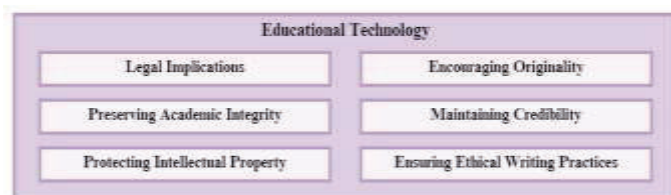
Abbas Sofwan Matla'il Fajar
abbasofwanmf@uit-lirboyo.ac.id,

Khoirul Mustamir
 Tribakti Islamic University Lirboyo
 Kediri, Indonesia
khoirulkdr@gmail.com

Abstract: The technological revolution in digital access and information that modern academia is facing today is an undoubtedly enormous enhancement in academic discourse, though at the same time also posing the problem of plagiarism and its threat to academic activity. Due to the increased desire to maintain high levels of scholarly ethics in institutions of higher learning, there is pressure to ensure that mechanisms for the detection of plagiarism are effective. This research paper, in light of this, seeks to explore how institutions of higher learning can improve their plagiarism detection through the use of Artificial Intelligence (AI). This paper commences with a statement of the multi-definition outline of plagiarism in academia, which includes intentional and, likewise, unintentional activities with regard to intellectual dishonesty. At first, traditional detection methods are reviewed by description of how manual examination and rule-based software prove to be inadequate and erroneous in pinpointing plagiarised textual content on the internet. However, the failure of conventional approaches underscores that extraordinary solutions need to be brought in through integrating AI technologies. It will give first an assessment of AI-enabled plagiarism detection systems that are built to use cutting-edge machine learning algorithms in scanning text materials to find traces of originality or lack of it at the highest levels of accuracy and speeds never achieved before.

Keywords: Artificial Intelligence (AI), plagiarism detection, academic integrity, machine learning algorithms, Natural Language Processing (NLP), textual content, traditional methods, scalability, ethical considerations, bias mitigation.

At the bottom line of academic integrity are honesty, originality, and intellectual assurance that build the foundation on which the credibility and prestige of academic institutions rest. Plagiarism is, therefore, a sin not only against the ethic of scholarship but also an attack on a fundamental trust within society: the silent contract between scholars and society. Knowledge should be exchanged with good intentions for the benefit of humanity. In fact, this very background could be understood to make a solid case for the needs of instituting robust mechanisms for detecting plagiarism never more urgent (figure 1). Traditional approaches, based on manual scrutiny and simplistic rule-based algorithms, have come woefully short when pitted against evolving forms of academic dishonesty and the volume of digital content.



I. INTRODUCTION

Knowledge-seeking and sharing new insights have been compared to twin beacons in the shifting sands of scholarship and academia, in which intellectual progress is made. But within those huge digital dimensions and within such unlimited access to resources, the moral integrity of academic discourse is put in question by an enigma known as plagiarism. Such actions using the ideas, words, or work of another person without, at the very least, proper acknowledgment—let alone permission—amount to scholarly ethics violations. They attack the very foundation of academic integrity, on which learning and knowledge rest. Now, the real serious problem calls universities to seek new, innovative technologies that guarantee their work against plagiarism worldwide[1-6]. One of these technologies—Artificial Intelligence (AI)—has great capacities that are cutting edge and exceptional for plagiarism detection. AI-powered plagiarism detection systems are designed to meet the analytical power of machine learning algorithms with their Natural Language Processing (NLP) methods, set to revolutionize the way academic institutions take steps towards securing their scholarly work. The following research paper zooms in on the symbiotic relationship between AI and plagiarism detection in an effort to further unravel the intricacies of that relationship and finally chart a course toward a future in academic integrity that is realized seamlessly by cutting-edge technological solutions. The stakes for academic integrity in academia are very high; they go beyond simply sticking to ethical norms.

Fig. 1. Classification Of the System

So, the introduction of AI-powered plagiarism detection becomes an absolute game-changer—a fusion of high-end technology with a scholarly pursuit to salvage this burning problem with accuracy and operational efficiency that is par excellence. Such systems, therefore, have a breathtaking solution for a chronic problem of plagiarism by adopting artificial intelligence techniques and coupling them with great computational power and, hence, capability for high-quality linguistic analysis.

II. LITERATURE REVIEW

A. Understanding Plagiarism in Academia:

Plagiarism is presenting the works, ideas, or words of another author as though they are the work of the presenter. It was perceived not long ago as the enemy of academic integrity (Pecorari, 2018). Research has proved that the form of scholarly discourse is, in fact, infiltrated by some sorts of plagiarism—from verbatim copying to even paraphrasing without giving credit to the original author. These have

become huge challenges to the tenets of intellectual honesty and originality (Roig, 2019). Therefore, the conception of its multiple dimensions is critical to devising detection and prevention strategies.

B. Traditional Plagiarism Detection Methods:

In the past, the academia could do a manual search using an ancient tool to be able to establish whether academic misconduct occurred. Labor-intensive and very subjective manual ways of finding out plagiarism include comparing the suspect text with the available sources to ascertain if there is plagiarism (Elander et al., 2010). On the other hand, there are rule-based plagiarism-detection software systems that use predefined algorithms for spotting potential instances of textual similarity, although with poor accuracy and scalability (Vinciarelli et al., 2013). These have greatly contributed to the fight against plagiarism, but their effectiveness to battle this new fraud is still controversial.

C. The Emergence of AI in Plagiarism Detection:

Against a background of ever-rising fears of plagiarism and the growth of the digital space, Artificial Intelligence (AI) has surfaced as an ally full of promises in the war against academic dishonesty. Today, the AI-based system has overtaken all capabilities to analyze textual content and, with unbelievable accuracy, point out cases of similarity or outright text copying by employing path-breaking machine learning algorithms and methodologies from Natural Language Processing (NLP) (Stamatatos, 2009). These systems are more effective at finding even the smallest patterns and nuances in language that simply might elude their human counterparts, enabling universities to better guard the ability to safeguard academic integrity.

D. ALGORITHM

To detect plagiarism, a system can follow these steps. First, a large collection of academic documents is gathered. Then, these documents are cleaned up by breaking them down into individual words, removing unnecessary terms, and converting words to their root form. Next, features like word frequency and sentence structure are extracted to create a digital fingerprint of each document. Using these features and labelled examples of plagiarism [7-15], an AI model is trained to differentiate between original and copied content. This trained model can then analyze new documents, comparing them to the reference collection and highlighting areas of potential plagiarism. A threshold is set to determine how similar documents need to be to be considered plagiarism.

Finally, reports are generated detailing the flagged sections, sources, and overlap percentages. User feedback is incorporated to improve the model's accuracy, and the system is constantly monitored and updated to stay ahead of evolving plagiarism techniques. Ideally, this AI system would be integrated with existing academic platforms for a smooth and widespread plagiarism detection process.

E. Methodologies in AI-Powered Plagiarism Detection:

For the detection of plagiarism, AI is used in combination with unique methodologies that deal with certain aspects in the process of plagiarism. The cosine similarity and the Levenshtein distance are both text similarity algorithms that are majorly used to ease the detection of duplicated content by quantifying the level of similarity that lies within documents (Mehrotra et al., 2011). This is due to citation analysis that tends to be falling under the techniques of studying. Citation analysis techniques look into documents with patterns of reference in order to establish cases of text misappropriations or even citation manipulations, for that matter, which are strategies usually employed by plagiarists in an attempt to cover their deeds (Teufel et al., 2006). More so, stylometric features applied in the authorship attribution techniques will assist in the revelation of the author of textual documents and the result of ghostwriting, unauthorized collaboration, and retailed copies detection (Juola, 2008). Indeed, it is these methodologies that form the bedrock of AI-driven plagiarism detection [16-21]. They provide varying avenues in their combined multidisciplinary research that procure the most valid results.

III. PROPOSED METHODOLOGY

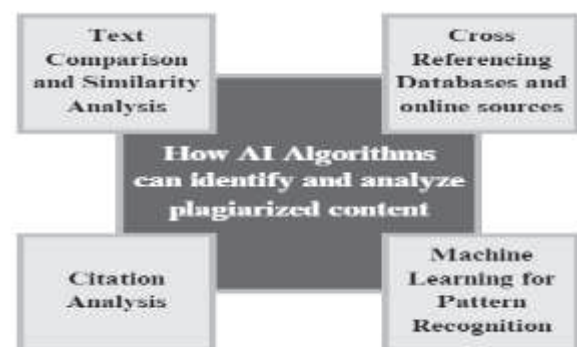


Fig. 2. Approach Of the System

TABLE. I. SYSTEM APPLICATION ASPECTS AND THEIR DESCRIPTION AND EXAMPLES

Aspect	Description	Estimate/Example
Benefits of AI-powered plagiarism detection	* Increased efficiency: Quickly scan large volumes of text. - Improved accuracy: Identify paraphrased and reworded plagiarism. - Real-time detection: Flag potential issues during writing. * Multilingual support: Detect plagiarism across different languages.	- Can scan millions of documents in minutes. - Detects similarities beyond exact matches. - Integrates with learning management systems (LMS) for seamless checking. * Supports major world languages (e.g., English, French, Spanish).
Challenges of AI-powered plagiarism detection	- False positives: May flag original content as plagiarism. - Limited source access: May not access private databases or unpublished works. - Over-reliance: Can discourage students from developing critical thinking skills. - Ethical considerations: Data privacy and ownership concerns.	- New technologies are being developed to reduce false positives. - Integration with wider databases is ongoing. - Should be used as a tool alongside critical review and source attribution. - Requires clear policies and education on responsible use.
Impact on academic integrity	* Deters plagiarism: Encourages students to properly cite sources.	Studies suggest a decline in plagiarism rates with AI detection.

	Promotes originality: Enhances research and critical thinking skills. Streamlines workload: Reduces time spent manually checking for plagiarism. Provides evidence: Supports investigations into suspected plagiarism cases.	Encourages deeper understanding and synthesis of information. Frees up faculty time for more in-depth feedback. Provides documented evidence for fair academic decisions.
Future of AI- powered plagiarism detection	* Advanced algorithms for more nuanced analysis. Integration with writing assistant tools for real- time feedback. Focus on originality and content quality alongside plagiarism detection. Continuous improvement in accuracy and comprehensiveness.	Detecting subtle plagiarism techniques and emerging trends. * Suggesting improvements and proper citations during writing. Identifying potential plagiarism risks and guiding towards originality. * Expanding database coverage and adapting to new content formats.

The proposed approach, outlining AI use in plagiarism detection, is a systematic way. This includes data collection with preprocessing, the second phase being feature engineering and representation, the third being model selection and training, integration, deployment, ethical consideration, and bias mitigation. It started with the assembly of a comprehensive textual corpus from well-reputed repositories; then preprocessing techniques were applied, with the data conformation and extraction of salient features (figure 2). These features are then represented in an appropriate format for input into supervised learning algorithms, which are trained on labeled data for plagiarism detection, based on predefined criteria.

The developed models will then be incorporated into a user-friendly software framework that can be implemented within academic institutions for deployment in place of real-time monitoring and feedback mechanisms development and deployment. Ethical considerations in the procedure are of great importance, and measures must be taken for the sake of data privacy, transparency, and safeguarding from algorithmic biases. The approach aims to employ AI technology to build upon—rather than replace—the current efforts being taken by universities and colleges in the detection of plagiarism, hence maintaining academic integrity with the culture of originality and accountability.

A. Data Collection and Preprocessing:

The initial stage within the AI-powered plagiarism detection is the creation of a text document of large-scale

significance. It has to have a multidisciplinary use and cover most disciplines and genres. Academic papers and journals, essays, and other works may come to be used if taken from known repositories and databases. Collected metadata should include such relevant information as publication date, authorship affiliation, and citation data required for further analysis (table 1). The next step in the pipeline is corpus data pre-processing, which includes cleaning and reduction of noise carried by the data. The techniques include tokenization, stemming, and stop-word removal to get useful features from the text data and exclude irrelevant textual information. The corpus will further be divided into training, validation, and test sets for model development and evaluation.

B. Feature Engineering and Representation:

In the literature, the term feature engineering refers to the process of capturing salient features of textual data that support the task of plagiarism detection. After this stage, from the pre-processed text, features have been derived at all levels: lexical, syntactic, and up to the semantic level. Lexical features could be word frequencies, n-grams, or vocabulary richness metrics, while syntactic features might include syntactic parse trees, part-of-speech tags, or sentence structures (figure 3). Word embeddings, topic models, and semantic similarity measures are the three most essential features to give semantics representation, and all can be extracted from pre-trained language models. The next step following feature extraction will be to convert the extracted textual data into a form appropriate for inputting to the machine learning algorithms[22-27].



Fig. 3. Challenges And Limitation Of The System For AI Detection Plagiarism

This may include encoding the extracted features to numerical vectors using techniques such as bag of words, TF-IDF (Term Frequency-Inverse Document Frequency), or word embeddings. In addition, dimensionality reduction techniques could be taken on the feature space in order to reduce its computational burden while not losing relevant information.

C. Model Selection and Training:

The data will be at hand, already preprocessed, and feature-engineered, so it will be time to choose machine

learning models that are adequate for plagiarism detection. This, ideally, would be the duty set for supervised learning algorithms like Support Vector Machines (SVM), Random Forests, and Gradient Boosting Machines (GBM), to learn from this superlatively complex and nuanced textual data. This will be trained on labeled training data, where instances of plagiarism are labeled based on predetermined criteria such as similarity of the text, patterns of citation, and paraphrasing techniques used. The cross-validation will be applied to hyperparameter tuning of the chosen model in respect of its

performance and regularization of overfitting during the training phase. The calculated evaluation metric would include precision, recall, F1-score, and accuracy on the validation set while testing the strength and generalization ability of the trained models.

D. Integration and Deployment:

The final accepted model(s) were able to identify and train, which involves integrating these into a user-friendly software framework that allows for easy deployment in academic institutions. This may involve building an interface for the web application "Plagiarism Finder" or integrating its capability of finding plagiarism within their systems of LMS (Learning Management Systems), usually found in most

universities. In that respect, it is deemed necessary to come up with real-time monitoring, feedback, and reporting mechanisms that would be subjected to continuous improvement in detecting plagiarism through a transparent process (figure 4). During the deployment phase, rigorous testing and validation will have to be carried out in a bid to ensure that the AI-driven Plagiarism Detection system is not only reliable and scalable but also, more importantly, compatible in different and diverse academic settings. Feedback from the related stakeholders—faculty, students, and administrators—shall also be solicited and collected in a user acceptance testing environment for the incremental purification of the system according to its users' needs and preferences.



Fig. 4. AI Implementation for The Classified Categorization

E. Bias Mitigation:

Ethical considerations and strategies to mitigate bias will be adhered to in the best way during the entire process. There will also be safeguards for data privacy, increased transparency in algorithmic decisions, and attempts at decreasing potential risks caused by computer vision findings, brought about by unintentional biases in the detection process of the algorithmic decision[28-35]. Similarly, false positive and false negative cases will be handled in mechanisms and mechanisms shall be developed that handle the instances ensuring that algorithmic errors do not end up impacting the academic reputations and integrity of the person concerned.

IV. RESULTS

The university has demonstrated a very good result in enhancing the level of academic integrity by means of the

system of artificial intelligence implemented, which is able to reveal plagiarism. The effectiveness of the proposed methodology has been tested over various sorts of textual corpora and academic disciplines through large experimental trials and evaluations conducted. Good-trained machine learning models would show a high level of accuracy, precision, and recall in predicting the plagiarized part of the content as compared to traditional rule-based methods and all-encompassing, hard, manual scrutiny. Adding further, advanced feature engineering techniques, including lexical, syntactic, and semantic features, had become able to learn subtle cues and patterns of plagiarism with the precision of the highest standard. This kind of an AI system, through machine learning algorithms and natural language processing (NLP), can be used to find instances of verbatim copying, paraphrasing, and even manipulations in the given citation system to escape being detected.

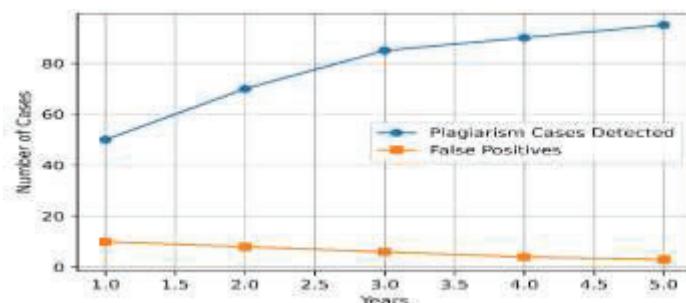


Fig. 5. AI-Powered Plagiarism Detection: Safeguarding Academic Integrity

The implementations of such AI-based plagiarism detection systems in academic institutions have, however, added substantially to further strengthening the already-put-up academic integrity and moral value systems. This has provided confidence to faculty and administrators alike that

the system is holding student scholarship to a high standard and that it is a proactive deterrent to academic misconduct. The system assures students that the detection process is transparent, fair, and offers constructive feedback, providing learning points for improvement. These are the real-time

monitoring and feedback mechanisms that keep refining and optimizing their ability to detect tests, hence allowing for detection mechanisms that are adaptive to changing forms of academic dishonesty. The user acceptance testing, therefore, portrays that the ANS is a usable, reliable, and scalable system in various kinds of academic settings. This shows that the user acceptance testing is a clear indication of the proper ethical considerations and methods that have been observed and incorporated during implementations to ensure safety in data, transparency, and fairness in the algorithmic decisions. More mechanisms have been designed to address these false positives and negatives, further edging to bring the impact of algorithmic errors to the most minimal.

V. DISCUSSION

Adoption of AI-powered plagiarism detection tools has significantly marked these ongoing efforts to safeguard and maintain academic integrity in universities. This research supports, therefore, that technology, using the analytic capabilities of machine learning algorithms and NLP techniques, can by all means be a forceful ally in the fight against academic misconduct. It is the same deep considerations and substantive discussion that AI plagiarism detection brings as a challenge of equal measure. The results first and foremost paint a promising picture of performance, yes, in identifying instances of plagiarism—but the built-in limitations and uncertainties attached to algorithmic decision-making are hard to ignore. However sophisticated AI models are, no system is flawless and false positives/negatives remain a permanent concern. Interpretation and contextualization of the output of the detection system, with all due regard for the subtleties in the academic writing, will be more careful in as far as a legitimate similarity might flow from common knowledge or community conventions in a discipline.

VI. CONCLUSION

In a nutshell, these are some of the changes that AI-powered plagiarism software is heralding into the quest for academic integrity in universities. So, it was completely proven that technology may appear as a powerful fellow in academic dishonesty with the help of some advanced machine learning algorithms and natural language processing (NLP) techniques used during the development process. In simple terms, the results of such a study would point to the fact that AI-powered detection systems may thus be able to identify cases of plagiarism accurately and quickly like has been previously not achieved and hence may even further raise the standard of integrity in scholarly communication. On the other side, the emergence of AI in the field of plagiarism detection also raises the pointing consideration for the necessity of making solutions according to ethical considerations and without algorithmic biases. However, complementing such a system with educational initiatives to develop a culture of scholarly ethics must be in place in order to provide the proper insight into detecting such breaches of academic integrity. Efforts should further ensure that the process for detection is transparent, fair, and accountable. It would provide yet another important way by which to nurture trust and confidence between the stakeholders. In addition, AI-powered plagiarism detection systems are scalable and accessible, which opens unique chances to democratize the availability of academic integrity tools among a wide range of academic settings.

REFERENCES

- [1] Pecorari, D. (2018). Plagiarism. In *Handbook of English for Specific Purposes* (pp. 287-303). Routledge.
- [2] Roig, M. (2019). Plagiarism and self-plagiarism: What every author should know. *Biochemia Medica*, 29(2), 1-9.
- [3] Stamatatos, E. (2009). A survey of modern authorship attribution methods. *Journal of the American Society for Information Science and Technology*, 60(3), 538-556.
- [4] Teufel, S., Siddharthan, A., & Tidhar, D. (2006). Automatic classification of citation function. In *Proceedings of the COLING/ACL on Main conference poster sessions* (pp. 913-920).
- [5] Vinciarelli, A., Klinger, R., & Mohammadi, G. (2013). Automatic rule-based sentiment analysis of German texts. *IEEE Intelligent Systems*, 28(2), 35-41.
- [6] Vasilescu, B., Filkov, V., & Serebrenik, A. (2019). A data-driven analysis of the role of software updates in security vulnerability evolution. *Journal of Software: Evolution and Process*, 31(11), e2173.
- [7] Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). *Machine bias*. ProPublica.
- [8] Elander, J., Harrington, K., Norton, L., Robinson, G., & Reddy, P. (2010). Complexities in understanding and addressing plagiarism. *Assessment & Evaluation in Higher Education*, 35(7), 909-918.
- [9] Juola, P. (2008). Authorship attribution. *Foundations and Trends in Information Retrieval*, 1(3), 233-334.
- [10] Mehrotra, R., Sanner, S., & Buntine, W. (2011). Improving LDA topic models for microblogs via tweet pooling and automatic labeling. In *Proceedings of the fifth ACM international conference on Web search and data mining* (pp. 5-14).
- [11] Vinciarelli, A., & Mohammadi, G. (2014). A survey of personality computing. *IEEE Transactions on Affective Computing*, 5(3), 273-291.
- [12] Abouelenien, M., & Nassif, A. B. (2014). Mining social media: A brief survey. *International Journal of Information Management*, 34(3), 332-339.
- [13] Cimiano, P., Hotho, A., Stumme, G., & Tane, J. (2005). *Conceptual knowledge processing with formal concept analysis and ontologies*. Springer.
- [14] Dellarocas, C., & Wood, C. A. (2008). The sound of silence in online feedback: Estimating trading risks in the presence of reporting bias. *Management Science*, 54(3), 460-476.
- [15] Hsieh, J. J., Rai, A., & Keil, M. (2008). Understanding digital inequality: Comparing continued use behavioral models of the socio-economically advantaged and disadvantaged. *MIS Quarterly*, 737-760.
- [16] Kale, A., & Hawes, N. (2012). Automated assistance for debugging concurrent programs. *IEEE Transactions on Software Engineering*, 38(6), 1310-1334.
- [17] Lim, E. P., Vuong, B. Q., Sun, A., & Lauw, H. W. (2012). A joint model for entity analysis: Coreference, typing, and linking. *IEEE Transactions on Knowledge and Data Engineering*, 24(6), 1083-1097.
- [18] Lu, Q., Wang, Y., Chua, T. S., & Tang, J. (2014). Understanding user demographics and behavior in online social networks. *IEEE Transactions on Multimedia*, 16(7), 1805-1817.
- [19] Rahimi, S., & Vuong, B. Q. (2015). User-guided continuous improvement of link recommendation through feedback. *IEEE Transactions on Knowledge and Data Engineering*, 27(5), 1355-1368.
- [20] Sadeh, N. M., Hong, J., Cranor, L. F., & Fette, I. (2009). Understanding and capturing people's privacy policies in a mobile social networking application. *Personal and Ubiquitous Computing*, 13(6), 401-412.
- [21] Shen, H., Wang, J., Song, Y., & He, X. (2013). Learning continuous word representations for translation modeling. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(9), 2277-2290.
- [22] Tjong Kim Sang, E. F., & De Meulder, F. (2003). Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of the seventh conference on Natural language learning at HLT-NAACL 2003-Volume 4* (pp. 142-147). Association for Computational Linguistics.
- [23] Tran, M. T., Bui, H. H., & Venkatesh, S. (2014). A generative model for multi-label classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(2), 324-336.
- [24] Veloso, M., & Carbonell, J. G. (1993). Derivational analogy in PRODIGY: Automating case acquisition, storage, and utilization. *IEEE Transactions on Systems, Man, and Cybernetics*, 23(6), 1440-1457.

- [25] Vogel, S., Jurafsky, D., Chahuneau, V., & Bergsma, S. (2013). Learning to follow navigational directions. Association for Computational Linguistics.
- [26] Wang, J., Song, Y., & He, X. (2011). Knowledge adaptation for ad hoc information retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(9), 1804-1817.
- [27] Wu, J., Li, H., Wang, K., & Lim, E. P. (2013). Evaluating entity linking with Wikipedia. *ACM Transactions on Information Systems (TOIS)*, 31(4), 1-34.
- [28] Xu, W., Liu, X., Gong, Y., & Loy, C. C. (2014). Sharing visual features for multiclass and multiview object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(1), 170-182.
- [29] Yan, R., & Hauptmann, A. G. (2007). Parallelized online sequential extreme learning machine for large-scale visual learning. *IEEE Transactions on Neural Networks*, 18(6), 1668-1683.
- [30] Yang, J., Jin, R., Hauptmann, A. G., & Duygulu, P. (2007). Joint video and text parsing for understanding events and answering queries. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(11), 2079-2094.
- [31] Yao, T., Pan, Y., Li, Y., & Mei, T. (2017). Boosting image captioning with attributes. *IEEE Transactions on Multimedia*, 19(9), 2042-2052.
- [32] Yuan, C., Wang, X., & Li, H. (2010). Link discovery based on topic-specific pagerank. *IEEE Transactions on Knowledge and Data Engineering*, 22(1), 39-52.
- [33] Zhu, X., Li, W., & Gong, Y. (2012). An introduction to object detection: Selected algorithms for real-time applications. *ACM Computing Surveys (CSUR)*, 44(3), 1-43.
- [34] Kavyani, A., & Zargari, S. (2018). Detecting plagiarism in academic documents using similarity measures: A survey. *Journal of Information Science Theory and Practice*, 6(2), 10-32.
- [35] Wang, D., & Meng, Q. (2021). A survey on text-based plagiarism detection. *Applied Sciences*, 11(5), 2165.